

AGNIBH DASGUPTA

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Education

University of Nebraska Omaha, Ph.D., Computing & Information Science	2021 – 2026
Utah State University, M.S., Computer Science	2018 – 2020
West Bengal University of Technology, B.Tech., Computer Science	2013 – 2017

Skills

ML Concepts:	Watermarking, Computer Vision, NLP, LLMs
Frameworks:	PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn
Programming:	Python, Bash, CUDA, Git, Linux
Distributed:	DeepSpeed, LoRA, PyTorch DDP, Slurm/HPC
Post-Training:	SFT, DPO, GRPO, TRL, RLHF
Agentic AI:	LangGraph, RAG, LLM-as-Judge, ChromaDB
Deployment:	FastAPI, vLLM, Quantization, OpenAI & Anthropic APIs

Professional Experience

University of Nebraska Omaha, Graduate Researcher	2021 – 2026
Representation learning, watermarking, and LLM research; four publications including CVPR 2026 and IEEE TAI 2025.	
Utah State University, Graduate Researcher	2018 – 2020
Deep reinforcement learning research for financial trading applications.	

Selected Research and Projects

Path of Exile Q&A Agent	2026
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- Wiki-grounded RAG for Path of Exile: live wiki retrieval with multi-query fusion and cross-encoder reranking
- LangGraph planner orchestrates multi-search as needed before synthesis; FastAPI backend and React trace UI
- Evaluated with labeled gold pages and LLM-as-judge scores for faithfulness and relevance

Invariant Features in LLMs: Geometric Characterization and Model Attribution (<i>under review</i>)	2026
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- Identified invariant subspaces in 9 open-source LLMs via contrastive sensitivity decomposition
- Validated semantic and nuisance components causally through next-token prediction interventions
- Achieved 90%+ zero-shot model attribution accuracy across base, fine-tuned, and distilled variants

TIACam: Text-Anchored Invariant Feature Learning with Auto-Augmentation	CVPR 2026
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- Built a zero-watermarking framework combining CLIP image-text features with a learned augmentation module
- Trained a differentiable auto-augmentor to find camera-like distortions, removing hand-designed augmentation families
- Achieved 95%+ watermark recovery on ImageNet and Visual Genome under real-world camera conditions

Watermarking Language Models through Language Models	IEEE TAI 2025
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- Built a model-agnostic watermarking scheme for LLM-generated text without touching model weights
- Tested on 25 open-source LLMs including fine-tuned and distilled variants using LoRA and DeepSpeed
- Achieved 88%+ detection accuracy under paraphrasing, fine-tuning, and adversarial prompt attacks

Robust Image Watermarking based on Cross-Attention and Invariant Domain Learning	CSCI 2023
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- Built a ViT-based cross-attention watermarking system with a self-supervised triplet loss objective
- Trained the encoder to cluster augmented views while separating different images in latent space
- Reached state-of-the-art comparable robustness under compound noise, confirmed via ablation

Perspective Transformation Layer	CSCI 2022
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- Proposed a lightweight learnable layer for perspective-robust recognition, replacing fixed pipelines
- Evaluated on SVHN, Imagenette, and MNIST under viewpoint variation with minimal extra compute
- Outperformed prior state-of-the-art on perspective benchmarks as a drop-in module